Stage 4 Technology

Farm in a Box: Grow More with Less | 25-period teaching program

Food and agricultural practices | agriculture-dominant network candidate v1.0

Course	Stage 4 Technology	Syllabus	Technology 7–8 Syllabus (2023)
Delivery	10 weeks | 25 x 60-minute periods	Pattern	5 periods per fortnight
Lesson balance	7 theory | 10 project/folio | 8 practical	Implementation	Current and mandatory from 2026
Official source	NSW Curriculum official course and FAP pages	Focus	Food and agricultural practices
Status	Network candidate | matching Word, PDF and manifest	v1.0	Teacher to confirm local delivery

This unit claims five outcomes and 19 of 21 Food and agricultural practices content items. Program rows use local audit IDs; the companion alignment manifest reproduces the exact official wording. Food-specific nutrition/preparation content is not claimed. The separate 55-period Agriculture elective remains unchanged.

Exact official Stage 4 outcomes

Outcome	Exact official wording	Outcome	Exact official wording
TE4-SDP-01	explains relationships between sustainability, design and production	**TE4-DES-01**	communicates and evaluates design ideas and solutions
TE4-PDP-01	describes the practices and processes of designers and producers	**TE4-SAF-01**	selects and safely uses tools, materials, technologies and processes
TE4-PPM-01	applies processes in the planning, management and production of projects		

What students are learning

Learning focus	In plain English	Learning focus	In plain English
Sustainability, design and production	Explain how crop-system decisions affect resources, production and sustainability.	**Design communication and evaluation**	Communicate ideas and judge the system against evidence-checkable criteria.
Designers and producers	Describe how agricultural producers and designers investigate, plan, establish, monitor and improve systems.	**Project planning and production**	Plan, manage, produce and document a portable crop-production project.
Safe practical work	Select and use only approved tools, materials, technologies and agricultural practices under current controls.		

Teacher-to-confirm register

When	Decision needed	Confirm by
Before the unit	Class needs, exact timetable, approved crop and viable growing location	Teacher / faculty
Before practical work	Materials, tools, workspace, supervision and current local risk controls	Teacher / faculty
Before Cultural content	Use the named public sources and confirm any additional Community/ICIP protocol	Teacher / faculty
Before formal assessment	Task authority, marks, dates, conditions, adjustments and submission method	Teacher / faculty

Teaching program | Periods 1–10

10 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
1	**Site/theory**	Agriculture as a connected system	Teach the three M1 sections. Map one familiar product as a system; compare the named NSW rice and cotton industries by product property, place, process and constraint.	FAP-ID-01, FAP-ID-02	Checks + system map + NSW industry comparison TE4-PDP-01, TE4-SDP-01
2	**Project/folio**	Launch: grow more with less	Unpack the challenge. Draft a problem, criteria and constraints without locking in a final solution.	FAP-ID-03, FAP-RP-08	Problem + criteria/constraints TE4-DES-01, TE4-PPM-01
3	**Site/theory**	First foods: plants, animals and nutrition	Use the named Wadawurrung, Gunditjmara and bounded Torres Strait Islander public sources to investigate plant and animal selection, use, nutrition and food availability without inferring local Cultural Knowledge.	FAP-RP-01	Attributed plant/animal comparison + protocol record TE4-SDP-01, TE4-PDP-01
4	**Project/folio**	Country and future farms	Compare the attributed Wadawurrung case with one emerging water-smart practice; state the comparison's limits.	FAP-RP-02	Sourced comparison + judgement TE4-SDP-01, TE4-DES-01
5	**Project/folio**	Sustainability lens	Apply environmental, social and production lenses; refine criteria and label supported, uncertain and local claims.	FAP-RP-08, FAP-PI-04	Lens + revised criteria TE4-SDP-01, TE4-DES-01, TE4-PPM-01
6	**Site/theory**	Plant parts and functions	Teach plant-part functions; link visible features to the needs of a small crop system.	FAP-ID-01	Plant annotation + response TE4-PDP-01
7	**Project/folio**	Crop and germination research	Research an approved crop and germination needs; justify a provisional choice against genuine constraints.	FAP-RP-03, FAP-RP-08	Sources + crop choice TE4-PDP-01, TE4-DES-01
8	**Practical**	Sow the approved crop	Under current controls, sow and label the approved crop; record starting conditions or use supplied evidence.	FAP-PI-01, FAP-PI-03	Observation + sowing record TE4-SAF-01, TE4-PPM-01
9	**Site/theory**	Growing media and circular nutrients	Teach media air, water, structure and nutrients; trace a circular compost pathway.	FAP-ID-01, FAP-ID-03	Media explanation + cycle TE4-PDP-01, TE4-SDP-01
10	**Practical**	Fair media test	Fair-test two approved media; record roles, limits, evidence and one improvement.	FAP-TE-01, FAP-TE-03	Fair-test record + recommendation TE4-SAF-01, TE4-PPM-01

Teacher to confirm before use: approved crop, materials, equipment, workspace, supervision, student adjustments and current risk controls for Periods 1–10. Cultural learning uses only the named, attributed public sources.

Teaching program | Periods 11–20

10 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
11	**Site/theory**	Water pathways and smart systems	Trace water movement and losses; compare one current or emerging water-smart technology.	FAP-RP-04, FAP-ID-03	Water map + technology comparison TE4-SDP-01, TE4-PDP-01
12	**Practical**	Build and test a wick	Build and measure an approved wick test; justify whether the concept is suitable.	FAP-PI-01, FAP-PI-02, FAP-PI-03	Wick data + suitability decision TE4-SAF-01, TE4-PPM-01
13	**Site/theory**	Read the microclimate	Map light, temperature, airflow and moisture patterns; explain one likely crop effect without invented precision.	FAP-ID-03, FAP-TE-03	Microclimate map + explanation TE4-PDP-01, TE4-SDP-01
14	**Site/theory**	Crop stress and biosecurity	Distinguish symptom from cause; practise quarantine, cleaning and reporting without handling unknown hazards.	FAP-RP-05, FAP-TE-03	Diagnosis + biosecurity response TE4-PDP-01, TE4-SDP-01
15	**Project/folio**	Brief, criteria and constraints	Finalise the brief, measurable criteria and genuine constraints for the portable crop system.	FAP-ID-03, FAP-RP-08	Approved brief + criteria TE4-DES-01, TE4-PPM-01
16	**Project/folio**	Concepts and decision matrix	Generate distinct concepts; annotate functions, compare with a matrix and refine the preferred direction.	FAP-RP-08, FAP-PI-04	Concepts + matrix + refinement TE4-DES-01, TE4-PPM-01
17	**Project/folio**	Production and risk plan	Plan the sequence, resources, quality checks and risk controls; justify choices and obtain approval.	FAP-TE-04, FAP-PI-03	Approved production/risk plan TE4-DES-01, TE4-PPM-01, TE4-SAF-01
18	**Practical**	Workspace readiness	Complete current induction/refresher and demonstrate stop, check and report routines before making.	FAP-PI-01, FAP-PI-03	Readiness checklist + observation TE4-SAF-01, TE4-PPM-01
19	**Practical**	Build and progressively test	Build in approved stages; test stability, water movement and access before irreversible steps.	FAP-PI-01, FAP-PI-02, FAP-PI-03	Progress + quality/modification record TE4-SAF-01, TE4-PPM-01
20	**Practical**	Establish the crop	Establish and label the approved crop; record starting conditions and a care routine, or use supplied data.	FAP-PI-01, FAP-PI-02, FAP-PI-03	System + care/start record TE4-SAF-01, TE4-PPM-01

Teacher to confirm before use: practical methods, construction materials, tool access, crop care, biosecurity response and current risk controls for Periods 11–20.

Teaching program | Periods 21–25

5 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
21	**Project/folio**	Useful field data	Build a consistent field record that separates measurements, observations and interpretation.	FAP-RP-04, FAP-RP-09	Field template + first data entry TE4-PPM-01, TE4-PDP-01
22	**Practical**	Maintain, test and improve	Carry out approved care; make one controlled change and retain failure or slow growth as valid data.	FAP-TE-01, FAP-TE-03	Before/after + retest record TE4-SAF-01, TE4-PPM-01
23	**Project/folio**	Community food and waste	Investigate a current community food initiative; propose one source-backed project connection without claiming a partnership.	FAP-RP-07, FAP-RP-05	Initiative profile + connection TE4-SDP-01, TE4-DES-01
24	**Practical**	Harvest, measure and trace quality	If authorised, harvest without consumption, measure and trace the crop; compare current lot records with emerging temperature/location sensors for distribution quality.	FAP-RP-03, FAP-RP-04, FAP-PI-01	Harvest/alternative + trace-technology comparison TE4-SAF-01, TE4-PPM-01, TE4-SDP-01
25	**Project/folio**	Judge quality and pitch version 2	Evaluate against criteria and pitch version 2: what to keep, change and test next.	FAP-RP-09, FAP-TE-05	Backed-up folio + evaluation/pitch TE4-DES-01, TE4-PPM-01, TE4-SDP-01
Teacher to confirm before use: harvest/non-harvest pathway, no-consumption boundary, final evidence expectations and any formal assessment decision for Periods 21–25.					

Evidence, feedback and support

Evidence	Look for	Respond / support
Theory and section checks	Accurate agricultural ideas and use of evidence	Return to the named section and model one stronger response
Design and folio	Criteria, planning, source records, decisions and refinements	Conference against one criterion and identify the next useful action
Practical observation	Safe setup, process, care, quality checks and clean-down	Pause, demonstrate and recheck before continuing
Crop/system result	Honest data and evaluation, including failure or slow growth	Use supplied comparison data when live growth is not viable